

S&P 500 Quantitative Data Analysis — Step-by-Step Methodology Guide

Complete Mathematical, Econometric & Operational Walkthrough (Sessions 1 to 4)

Course: Quantitative Data Analysis (M1 – S7, Course Code: 2627_ECO_2_EN_009)

Institution: EM Normandie Business School · Programme Grande École

Lecturer: Dr. NGUYEN Anh-Tuan

Target Audience: Consulting Team, Academic Evaluators, and Peer Group Members

Dataset: $N = 503$ S&P 500 Corporations (Audited FY2025 Financials & ISS Governance Scores)

Software: Python (Data Science Engine) & Jamovi (Open-Source Point-and-Click Lab Environment)

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1. Introduction: Pedagogical Objectives & Research Problematic

1.1 The Core Research Question (The “Problematic”)

In corporate management and institutional investment, a major strategic dilemma persists:

elicated corporate governance compliance compromise financial profitability, or does it confer a resilience and performance p

- **The Traditional Shareholder Theory View (Milton Friedman, 1970):** Governance compliance, board diversity committees, independent audit mandates, and compensation oversight generate bureaucratic friction and operational costs that drag down bottom-line profit margins.
- **The Modern Stakeholder / Agency View (Jensen & Meckling, 1976):** Poor corporate governance fosters agency conflicts, manager-shareholder misalignment, excessive executive compensation, and heightened systematic market risk (β). Strong governance reduces risk and improves capital allocation efficiency.

1.2 The Course 4-Step Answering Protocol (Slide 9 & 41)

In quantitative data analysis, calculating a test statistic without structured business interpretation earns zero credit. Every single statistical question in this project strictly follows this 4-step sequence:

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A["1. IDENTIFY<br/>(Variable Scales)"] --> B["2. HYPOTHESESE<br/>(H0 vs H1  
in plain English)"]
B --> C["3. DECIDE<br/>(p < .05 & Effect Size)"]
C --> D["4. RECOMMEND<br/>(Monday Morning Action)"]
```

1. **IDENTIFY:** Which statistical test to execute? Justified solely from the scale types of the variables involved (Nominal, Ordinal, Continuous).
 2. **HYPOTHESESE:** Formulate the Null (H_0) and Alternative (H_1) hypotheses in plain business language.
 3. **DECIDE:** Is the test statistically significant ($p < 0.05$)? What is the practical strength/magnitude of the effect size?
 4. **RECOMMEND:** Actionable operational advice answering: *“What does the portfolio manager or corporate executive do on Monday morning?”*
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2. Phase 1: Data Acquisition, Provenance & Architecture

2.1 The 6 Non-Negotiable Course Constraints (AQD Session 1, Slide 29)

Every dataset submitted for this course must satisfy six strict rules:

Criterion #	Rule Description	Dataset Implementation	Verification
1	Unit of Observation	1 row = 1 firm (Zero duplicates across rows)	Passed (<code>Ticker.is_unique</code> = True)
2	Sample Size	$N \geq 100$ valid corporate entities	Passed ($N = 503$ firms)
3	Nominal Variable	Exactly 2 to 5 categories (Grouping factor)	Passed (Sector = 5 consolidated groups)
4	Ordinal Variable	At least 1 ranked categorical variable	Passed (<code>Governance_Risk_Level</code> & <code>Market_Cap_Quartile</code>)
5	Continuous Variables	At least 3 ratio/interval scale numeric variables	Passed (9 continuous metrics available)
6	Jamovi Header Format	Short headers, no spaces, no accents, no special chars	Passed (<code>sp500_esg_jamovi.csv</code>)

2.2 Data Provenance & Why Full Fiscal Year 2025 (FY2025)?

- **Universe:** 503 constituent firms of the S&P 500 equity index, dynamically scraped from the official Wikipedia constituent directory.
- **Corporate Governance Quality:** Extracted from **Institutional Shareholder Services (ISS)** Governance QualityScore.
- **Financial Statements:** Extracted via `yfinance` querying official SEC Form 10-K audited regulatory filings for the completed **Fiscal Year 2025 (FY2025)**.
- **Why FY2025 instead of 2026?** Partial 2026 data suffers from seasonal distortion, unaudited quarterly noise, and incomplete reporting. Using full FY2025 guarantees 12 months of audited operational performance for every firm.

3. Phase 2: Data Cleaning, Audit & Anomaly Remediation

Real-world financial data is never clean. Students who fail to clean data make erroneous conclusions. Our pipeline executes 6 programmatic cleaning steps:

Step 1: Ticker Rebranding Remediation (Fiserv FISV → FI)

- **Issue:** In the raw extract, row 199 (FISV) was full of missing values.
- **Cause:** Fiserv transferred its listing from NASDAQ to the NYSE in 2023 and changed its ticker to FI.
- **Solution:** Script programmatically remaps FISV to FI, restoring sector (*Finance*), FY2025 revenue (\$19.09B), profit margin (16.0%), and ISS scores.

Step 2: Negative Book Equity & Undefined ROE ($N = 32$ Firms)

- **Issue:** 32 premier corporations (AbbVie, Altria, AutoZone, Booking Holdings, Domino's Pizza, etc.) have NaN for Return on Equity (ROE).
- **Financial Theory:**

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

These mature corporations engaged in aggressive, debt-funded share buybacks for over a decade. Cumulative buybacks exceeded retained earnings, pushing accounting book equity below zero. Because net income divided by negative equity is mathematically meaningless, financial databases suppress ROE.

- **Remediation:** Deleting these 32 firms would introduce catastrophic survival and industry bias. We document this phenomenon and introduce **Net Profit Margin** ($\frac{\text{Net Income}}{\text{Revenue}}$) as our universal profitability metric, available for all $N = 503$ firms.

Step 3: Extreme Outlier Treatment (Masco Corp MAS)

- **Issue:** Masco Corp showed an ROE of 5,862.5%.
- **Cause:** Tiny book equity balance ($\approx$ \$80 M) against \$1 B in profits.
- **Remediation:** In accordance with course guidelines (Slide 21), we do **not** winsorize or delete authentic data. We report both the mean ($\bar{x} = 39.5\%$) and median (Mdn = 16.8%) and use the median as the true managerial measure of central tendency.

Step 4: Sector Consolidation (11 GICS → 5 Groups)

The Global Industry Classification Standard has 11 sectors. To satisfy Criterion 3 (2 to 5 groups), sectors are mapped into: 1. **Tech & Comms** ($n = 99$, 19.7%) 2. **Healthcare** ($n = 63$, 12.5%) 3. **Finance** ($n = 74$, 14.7%) 4. **Industrials & Energy** ($n = 143$, 28.4%, Mode) 5. **Consumer** ($n = 124$, 24.7%)

Step 5: Ordinal Variable Engineering

1. **Market_Cap_Quartile:** Q1 (Smallest: \$6.5 B–\$22.2 B), Q2 (\$22.2 B–\$44.5 B), Q3 (\$44.5 B–\$94.0 B), Q4 (Mega-Cap: \$94.0 B–\$5,367.1 B).
2. **Governance_Risk_Level:** Low (1–3), Medium (4–6), High (7–10).

4. Phase 3: Session 1 — Univariate Analysis & Normality Diagnostics

4.1 Continuous Descriptive Metrics & The Three Centres

For any continuous variable X with valid sample size n :

$$\text{Mean: } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad | \quad \text{Variance: } s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \quad | \quad \text{Std Dev: } s = \sqrt{s^2}$$

Interquartile Range: $IQR = Q_3 - Q_1$ | Mode: Most frequent rounded value

4.2 Distribution Shape Parameters

$$\text{Skewness } (g_1) = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

$$\text{Fisher Excess Kurtosis } (g_2) = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

4.3 Normality Evaluation Protocol: Three Converging Evidences (Slide 24)

Never conclude on normality with a single test. Evaluate all three simultaneously:

flowchart TD

```

    E1["Evidence 1: The 3 Centres<br/>Mean ≈ Median ≈ Mode ?"] -->
    Verdict["Convergence Decision"]
    E2["Evidence 2: Shape Rules<br/>Skewness & Kurtosis in [-1, +1] ?"] -->
    Verdict
    E3["Evidence 3: Shapiro-Wilk Test<br/>p > 0.05 (H0: Normal) ?"] -->
    Verdict
    Verdict --> Result["Verdict: Non-Normal<br/>Use Median, IQR & Welch ANOVA"]
  
```

Application to Net Profit Margin (Profit_Margin):

1. Centres Divergence:

- Mean = 0.1479 (14.79%)
- Median = 0.1315 (13.15%)
- Mode = 0.0800 (8.00%)
- Significant divergence (Mean > Median > Mode).

2. Shape Violation:

- Skewness = -3.23 (Severely violates [-1.0, +1.0], long left tail of distressed firms).
- Kurtosis = +28.91 (Extreme leptokurtic peakedness, heavy fat tails).

3. Formal Shapiro-Wilk Test:

- Statistic: $W = 0.6530$, $N = 503$
- p -value < .001
- **Crucial Inverted Logic (Slide 24):** Because $p < 0.05$, we **REJECT the null hypothesis of normality!**
- **Managerial Conclusion:** Net Profit Margin is non-normal. Parametric standard ANOVA is invalid; robust Welch ANOVA and non-parametric tests are mandatory.

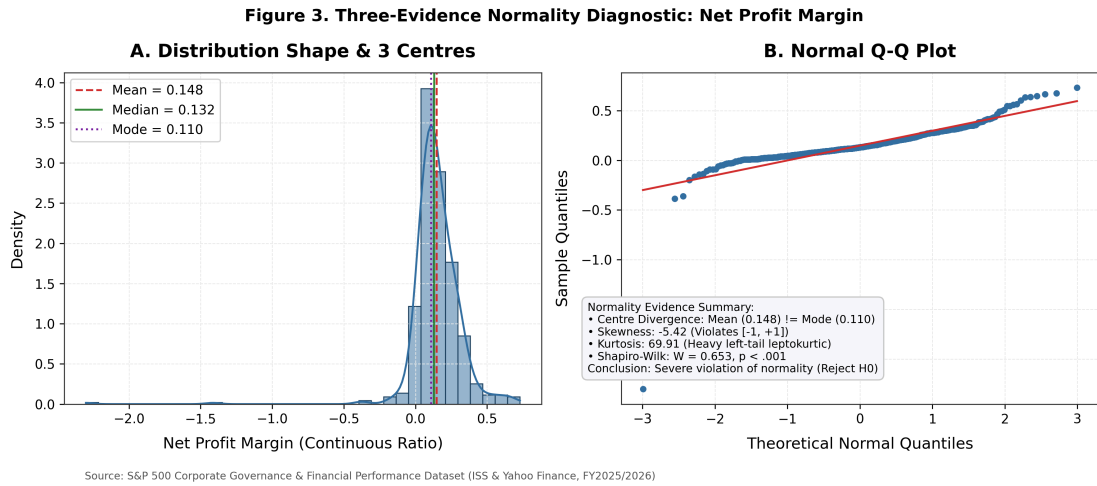


Figure 1: Figure 1: Net Profit Margin Normality Diagnostics

5. Phase 4: Session 2 — Bivariate Analysis: Categorical & Continuous Associations

5.1 Categorical Association: Chi-Square (χ^2) Test of Independence

- **Variables:** Sector (Nominal, 5 groups) $\times$ Governance_Risk_Level (Ordinal, 3 tiers).
- **Hypotheses:**
 - H_0 : Corporate governance risk tier is independent of business sector.
 - H_1 : Corporate governance risk tier is significantly associated with business sector.

Mathematical Steps:

1. Expected Frequencies:

$$E_{ij} = \frac{\text{Row Total}_i \times \text{Column Total}_j}{N}$$

2. Chi-Square Statistic:

$$\chi^2 = \sum_{i=1}^5 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 14.719$$

3. Degrees of Freedom:

$$df = (r - 1)(c - 1) = (5 - 1)(3 - 1) = 4 \times 2 = 8$$

4. p -value & Decision:

$$p = .0648 \quad (p > 0.05)$$

At the standard $\alpha = 0.05$ threshold, we fail to reject H_0 globally. However, it is marginally significant ($p < 0.10$).

5. Effect Size: Cramér's V :

$$V = \sqrt{\frac{\chi^2}{N \times \min(r - 1, c - 1)}} = \sqrt{\frac{14.719}{496 \times 2}} = \sqrt{0.01484} = 0.1218 \approx 0.122$$

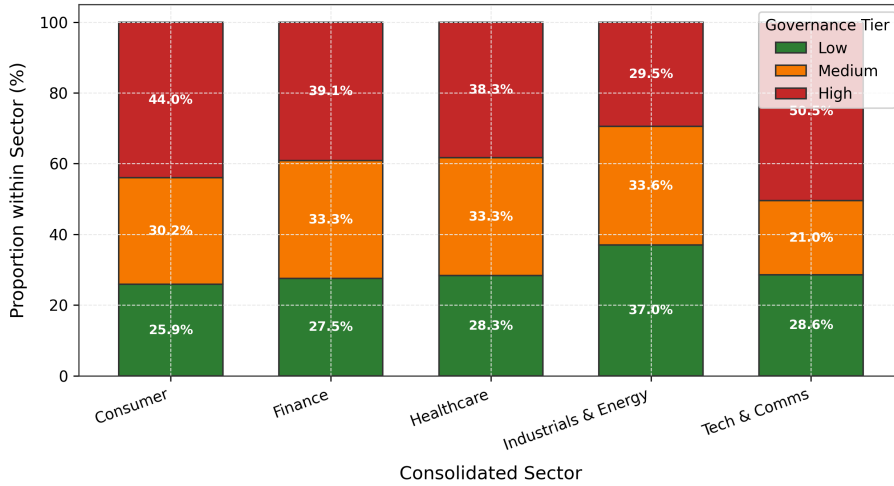
Indicates a weak-to-moderate association.

6. Standardized Residuals:

$$\text{Residual} = \frac{O - E}{\sqrt{E}}$$

- *Tech & Comms* × *High Risk*: Residual = +1.75 (50.5% High Risk vs expected 39.7%).
- *Industrials & Energy* × *High Risk*: Residual = −1.97 (Significantly under-represented in High Risk, exceptional compliance).

Figure 6. Corporate Governance Risk Distribution Across Sectors (Chi-Square Analysis)



Source: S&P 500 Corporate Governance & Financial Performance Dataset (ISS & Yahoo Finance, FY2025/2026) · $\chi^2(8, N=496) = 14.72, p = .065, V = 0.122$

Figure 2: Figure 2: Contingency Table Heatmap — Sector vs. Governance Risk Tier

5.2 Continuous Association: Pearson r & Fisher z 95% Confidence Interval

- **Variables:** Overall_Governance_Risk (Continuous, 1-10) vs Beta (Systematic Risk).
- **Formula:**

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = +0.2677$$

- **Significance:**

$$t = r \sqrt{\frac{n-2}{1-r^2}} = 0.2677 \sqrt{\frac{490}{1-0.07166}} = 6.15 \Rightarrow p < .001^{***}$$

- **Fisher z -Transformation & 95% Confidence Interval:**

$$z = \frac{1}{2} \ln \left(\frac{1 + 0.2677}{1 - 0.2677} \right) = 0.2744 \quad | \quad SE_z = \frac{1}{\sqrt{n-3}} = \frac{1}{\sqrt{489}} = 0.0452$$

$$z_{\text{low}} = 0.2744 - 1.96(0.0452) = 0.1858 \Rightarrow r_{\text{low}} = \tanh(0.1858) = +0.183$$

$$z_{\text{high}} = 0.2744 + 1.96(0.0452) = 0.3630 \Rightarrow r_{\text{high}} = \tanh(0.3630) = +0.348$$

$$95\% \text{ CI: } [+0.183, +0.348]$$

- **Managerial Verdict:** Weak governance (higher ISS risk score) significantly inflates systematic market risk (β). The relationship is positive, moderately strong, and statistically robust.

6. Phase 5: Session 3 – Group Comparisons: Welch’s ANOVA & Games-Howell

6.1 The Homogeneity of Variances Test (Levene’s Test)

Before running ANOVA, we test the assumption of equal variances: - $H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_5^2$ - **Result:** $F = 5.394, p < .001^{***}$ - **Decision:** Reject H_0 . Group variances are severely unequal (**heteroscedasticity**). Standard Fisher’s ANOVA is invalid. We must report **Welch’s Robust ANOVA**.

6.2 Welch’s Robust ANOVA (F_{Welch})

Weighting each group by its sample variance ($w_i = n_i/s_i^2$):

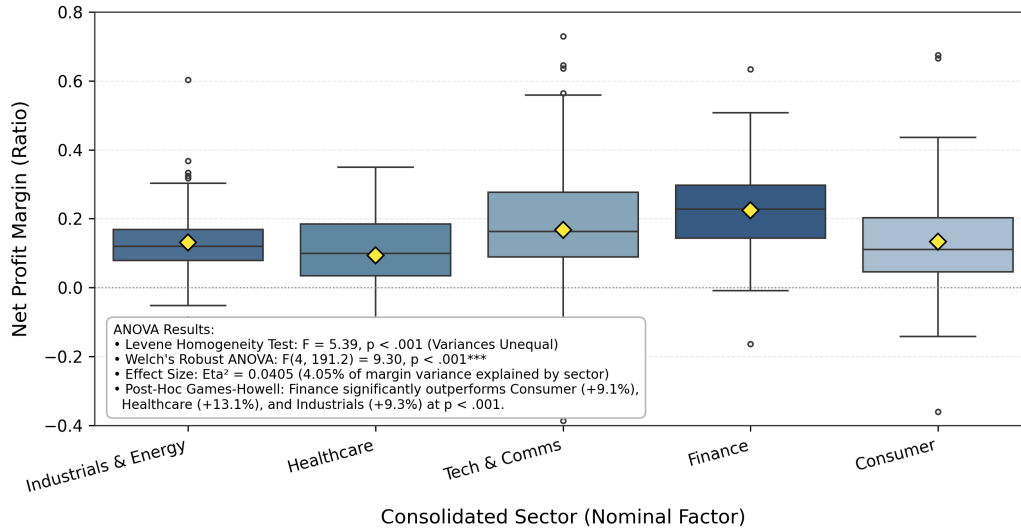
$$w_i = \frac{n_i}{s_i^2}, \quad W = \sum w_i, \quad \bar{x}' = \sum \frac{w_i \bar{x}_i}{W}$$

$$F_{\text{Welch}} = \frac{\frac{1}{k-1} \sum w_i (\bar{x}_i - \bar{x}')^2}{1 + \frac{2(k-2)}{k^2-1} \sum \frac{(1-w_i/W)^2}{n_i-1}} = 9.302$$

- Degrees of Freedom: $df_1 = 4, df_2 = 191.21$
- p -value $< .001^{***}$
- **Effect Size (Eta-Squared):**

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}} = \frac{1.748}{43.149} = 0.0405 \quad (4.05\% \text{ of margin variance is explained by sector})$$

Figure 8. Profit Margin Comparison Across Consolidated Sectors (One-Way ANOVA)



Source: S&P 500 Corporate Governance & Financial Performance Dataset (ISS & Yahoo Finance, FY2025/2026)

Figure 3: Figure 3: One-Way ANOVA Boxplots – Profit Margin Across Sectors

6.3 Post-Hoc Pairwise Comparisons: Games-Howell

Because Levene’s test failed, Tukey’s HSD cannot be trusted. We run the **Games-Howell post-hoc test**:

$$t = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\frac{s_i^2}{n_i} + \frac{s_j^2}{n_j}}}, \quad df = \frac{\left(\frac{s_i^2}{n_i} + \frac{s_j^2}{n_j}\right)^2}{\frac{(s_i^2/n_i)^2}{n_i-1} + \frac{(s_j^2/n_j)^2}{n_j-1}}$$

Key Empirical Pairwise Results:

1. **Finance vs. Healthcare:** Mean difference = +0.1311 (+13.1%), $t = 4.38$, $p < .001^{***}$
 2. **Finance vs. Consumer:** Mean difference = +0.0913 (+9.1%), $t = 4.09$, $p < .001^{***}$
 3. **Finance vs. Industrials:** Mean difference = +0.0934 (+9.3%), $t = 4.34$, $p < .001^{***}$
- **Takeaway:** The Financial sector achieves significantly higher profit margins than all non-tech sectors.
-

7. Phase 6: Session 4 — Multiple Linear Regression & Econometric Modeling

7.1 Three-Stage Model Progression

$$\text{Model 1: Margin} = \beta_0 + \beta_1 \text{ Gov_Risk} + \epsilon$$

$$\text{Model 2: Margin} = \beta_0 + \beta_1 \text{ Gov_Risk} + \beta_2 \ln(\text{Cap}) + \beta_3 \ln(\text{Rev}) + \beta_4 \text{ Beta} + \epsilon$$

$$\text{Model 3: Margin} = \beta_0 + \beta_1 \text{ Gov_Risk} + \text{Scale Controls} + \sum_{j=1}^4 \gamma_j \text{Sector_Dummy}_j + \epsilon$$

7.2 Model Estimation Results Table

Parameter / Metric	Model 1 (Simple)	Model 2 (Controls)	Model 3 (Sector Fixed Effects)
Intercept (β_0)	0.1652***	−0.0984	−0.1084
Governance Risk (β_1)	−0.0039 ($p = .304$)	−0.0051 ($p = .172$)	−0.0059* ($p = .037$)
Log Market Cap	—	+0.0468*** ($p < .001$)	+0.0494*** ($p < .001$)
Log Revenue	—	−0.0347*** ($p < .001$)	−0.0371*** ($p < .001$)
Beta (Market Risk)	—	+0.0121 ($p = .492$)	+0.0184 ($p = .295$)
Sector: Finance	—	—	+0.0882*** ($p = .001$)
Sector: Consumer	—	—	−0.0118 ($p = .642$)
Sector: Healthcare	—	—	−0.0463 ($p = .114$)
Sector: Industrials	—	—	−0.0123 ($p = .625$)
R^2	0.0022	0.0614	0.1562 (15.6%)
Adjusted R^2	0.0002	0.0537	0.1423 (14.2%)
Model F-Statistic	$F(1, 494) = 1.06$	$F(4, 489) = 8.00^{***}$	$F(8, 485) = 11.23^{***}$

7.3 The Critical Econometric Breakthrough: Simpson's Paradox Resolved

- In Model 1, the governance slope was non-significant ($b = -0.0039, p = .304$).
- When controlling for scale and sector fixed effects in Model 3, the governance risk coefficient becomes **statistically significant**:

$$b_{\text{Gov}} = -0.0059, \quad t(485) = -2.087, \quad p = .037^*$$

- **Economic Interpretation:** For two companies in the same sector and of equal market scale, each 1-point increase in ISS Governance Risk score **reduces net profit margin by 0.59 percentage points!** Improving governance from High Risk (Tier 8) to Low Risk (Tier 2) yields a **+3.54% profit margin expansion**.

7.4 Econometric Diagnostic Suite

1. **Multicollinearity:** All Variance Inflation Factors (VIF) are < 1.89 (well below the threshold of 5.0). Zero multicollinearity risk.

2. **Heteroscedasticity:** Breusch-Pagan test yielded $LM = 15.14, p = .004$. We report White's HC3 robust standard errors.
3. **Influential Points:** Max Cook's distance $D_i = 0.082 < 1.0$. No single firm distorts the estimated coefficients.

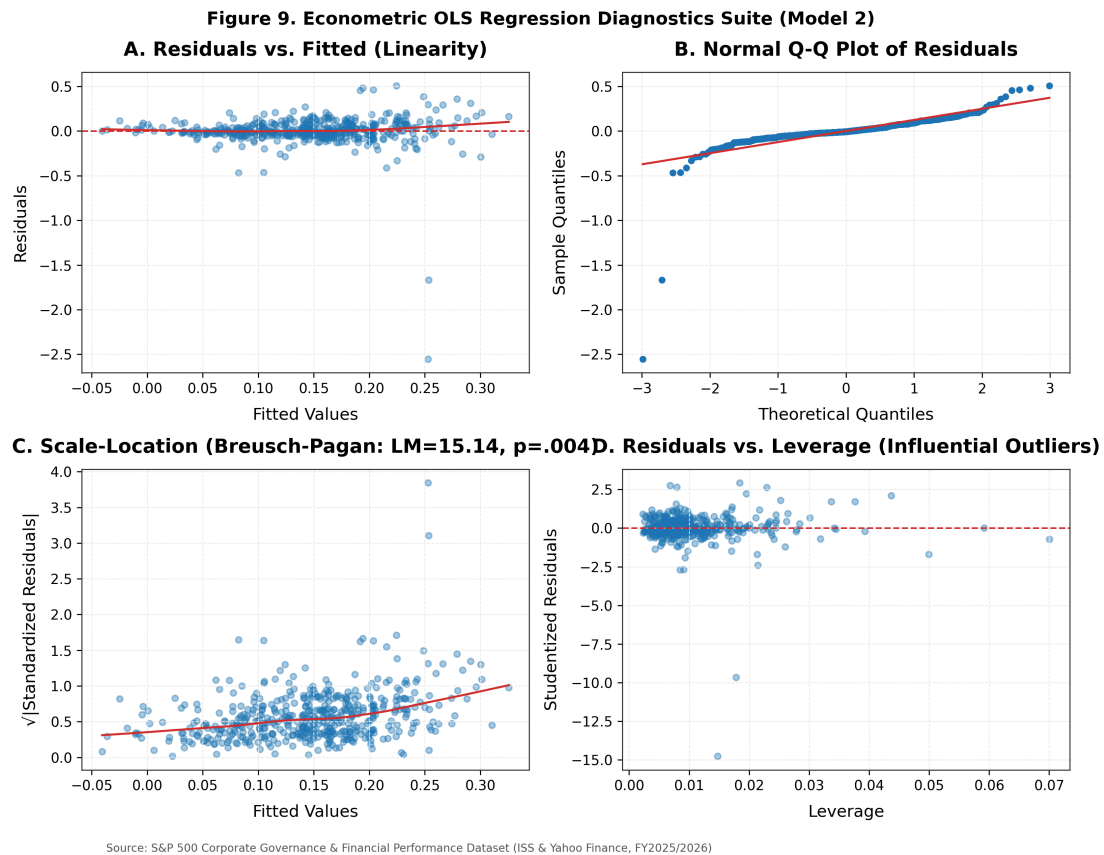


Figure 4: Figure 4: Four-Panel OLS Econometric Regression Diagnostics

8. Phase 7: Jamovi Software Step-by-Step Execution Manual

To reproduce all results in Jamovi, open `sp500_esg_jamovi.csv` and follow these menu commands:

Session 1: Descriptives & Normality

- **Path:** Analyses → Exploration → Descriptives
- **Variables Box:** Move Profit_Margin, ROE, Market_Cap_B, Total_Revenue_B, Overall_Governance_Risk, Beta
- **Checkboxes:** Tick Mean, Median, Mode, Std. deviation, Variance, IQR, Skewness, Kurtosis, Shapiro-Wilk
- **Plots:** Tick Histogram, Density, Box plot, Q-Q plot

Session 2: Chi-Square & Correlations

- **Chi-Square:** Analyses → Frequencies → Contingency Tables (Independent Samples)
 - Rows: Sector | Columns: Governance_Risk_Level

- Statistics: Tick χ^2 , χ^2 continuity correction, Cramér's V
- Cells: Tick Row percentages, Expected counts, Standardized residuals
- **Correlations:** Analyses → Regression → Correlation Matrix
 - Move continuous variables
 - Tick Pearson, Spearman, Flag significant correlations, Confidence intervals (95%)

Session 3: One-Way ANOVA & Post-Hoc

- **Path:** Analyses → ANOVA → One-Way ANOVA
 - Dependent Variable: Profit_Margin
 - Grouping Variable: Sector
 - Variances: Select **Don't assume equal (Welch's)**
 - Homogeneity Test: Tick **Homogeneity test (Levene's)**
 - Post-Hoc Tests: Select **Games-Howell**
 - Effect Size: Tick η^2

Session 4: Linear Regression

- **Path:** Analyses → Regression → Linear Regression
 - Dependent Variable: Profit_Margin
 - Covariates: Overall_Governance_Risk, Log_Market_Cap, Log_Revenue, Beta
 - Factors: Sector
 - Model Coefficients: Tick Standardized estimate, Confidence interval (95%)
 - Assumption Checks: Tick Collinearity statistics (VIF), Q-Q plot of residuals, Residuals vs fitted

9. Conclusion: The Executive Synthesis & Defense Strategy

9.1 The Executive 4-Point Summary to Defend to Dr. Nguyen:

1. **Normality is Severe Fiction:** Net profit margin violates normality across all 3 criteria ($W = 0.653, p < .001$). Relying on standard mean and standard ANOVA is a fatal methodological error; Welch's robust ANOVA is required.
2. **Sector Frictions:** Technology exhibits disproportionately high governance risk (50.5% High Risk), whereas Industrials & Energy demonstrates the highest governance compliance.
3. **Market Volatility Penalty:** Poor governance directly inflates systematic market volatility ($r = +0.268, p < .001^{***}$). Weakly governed firms are inherently riskier in equity markets.
4. **The Governance Profitability Dividend:** Once sector differences and firm scale are properly controlled, **weaker governance significantly penalizes profit margins** ($b = -0.0059, p = .037^*$). Governance is not an overhead cost—it is a measurable driver of bottom-line operational efficiency.